DAILY AI BRIEFING · LEARN ONE THING + WHAT MOVED

Keeping up with the AIs

Big teacher, small student

Issue #31 | Fri, 30 Oct 2026 | ☕ ~9 min read

⚡ If you only have one minute

1. **Mistral finally released the weights for Large 4**, a roughly 1-trillion-parameter model. Free to download and build on, with **one catch for API resellers**.
2. **Researchers broke the built-in browsers of five AI coding tools.** Fixes have shipped for Cursor and Copilot, but **Codex and VS Code are still unpatched**.
3. **Part 1 is distillation:** how a big model teaches a small one, and why the small one can sometimes **beat its teacher on one job**.

Today in three numbers

1.06 TB
download size of **Mistral Large 4's** open weights

2 of 5
AI coding tools **still unpatched** after the EMBRACE research

5
core values in OpenAI's **rewritten Model Spec**

PART 1 Learn one thing

🧑‍🏫 Distillation: how big models teach small ones

4 MIN READ | MODULE 4 · SHIPPING FOR REAL

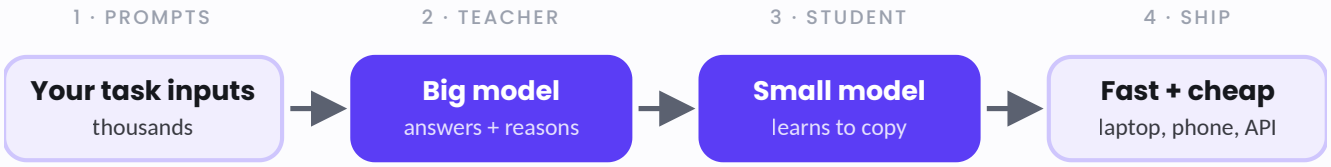
The cheapest way to get a small model that's good at your job is to let a big one teach it.

Distillation means training a small "student" model to copy a big "teacher" model. The teacher answers thousands of prompts. The student learns from **those answers, not just the original data**, so it picks up the teacher's habits in a much smaller package.

A **small language model (SLM)** is roughly anything you can run on a laptop, a phone or one cheap GPU: think **under 10 billion parameters**. Distillation is one of the main ways labs build them.

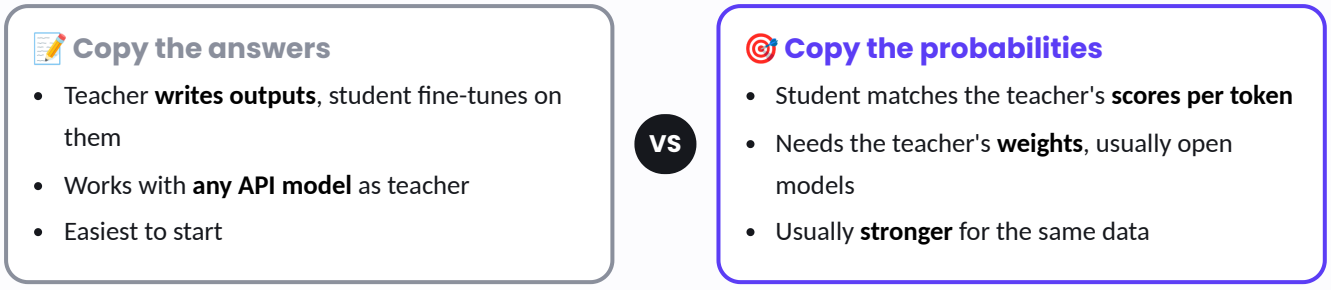
Analogy: a senior chef writing down every dish for a new cook. The new cook doesn't need 30 years in the kitchen. They study the chef's actual plates, and the reasons behind them. They'll never match the chef on everything, but on **the ten dishes your dhaba serves**, they can get very close, and cost far less per shift.

How distillation works



Simplified. Some methods copy only the teacher's final text; others copy its full probability scores for every token.

Two flavours. The simple one: have the teacher **write answers**, then fine-tune the student on them (Issue #30). The stronger one copies the teacher's **probabilities**: not just "the next word is *blue*", but "80% blue, 15% grey". That extra signal teaches the student how confident to be. NVIDIA's guide describes this as copying the teacher's **logits** (raw scores). (NVIDIA)



Both are common. Copying probabilities usually needs teacher and student to share a vocabulary (MaxText docs).

How small can you go? In Google's "distilling step-by-step" research, the teacher wrote out **its reasoning as well as its answers**. A **770M-parameter** student then beat a few-shot **540B PaLM** on a benchmark, using only **80% of the training examples**: over **700x smaller**. (Google Research)

The catch: a student is great at **what it was taught** and weaker everywhere else. That's why it's worth it for one narrow, high-volume job, not a general chatbot.

Why now? Mistral says **smaller distilled versions** of yesterday's 1T model are coming "in the coming months", **including one for laptops**. (TNW) Distillation is how frontier models trickle down to hardware you own.

📖 **Jargon buster**

- **Teacher / student:** the **big model** that knows the task and the **small one** learning from it.
- **Logits:** the model's **raw scores for every possible next token**, before they become probabilities.
- **KL divergence:** a number for **how different two probability lists are**. Distillation tries to shrink it.
- **Pruning: cutting layers or neurons** from a model, often followed by distillation to recover quality.

🧪 Try it in 10 minutes

- **No code:** pick a task you repeat. Ask a big model to answer **20 examples, explaining its reasoning**. Paste 5 of those into a small, cheap model's prompt as examples and test the rest. If it does well, you've just found **a job a student could learn**.
- **Some code:** run the TRL **GKD trainer** example below on a free Colab GPU. It uses a 1.5B teacher and a 0.5B student, both small enough to fit.

🛠️ For devs: distillation in TRL

```
from trl import GKDConfig, GKDTrainer
from transformers import AutoModelForCausalLM, AutoTokenizer
tok = AutoTokenizer.from_pretrained("Qwen/Qwen2-0.5B-Instruct")
student = AutoModelForCausalLM.from_pretrained("Qwen/Qwen2-0.5B-Instruct")
teacher = AutoModelForCausalLM.from_pretrained("Qwen/Qwen2-1.5B-Instruct")
args = GKDConfig(output_dir="gkd-model", per_device_train_batch_size=1)
GKDTrainer(model=student, teacher_model=teacher, args=args,
           processing_class=tok, train_dataset=ds).train()  # ds: chat "messages"
```

Gotchas: pick teacher and student from **the same family** so their tokenizers match. TRL's `lmbda` setting controls how much the student learns from **its own attempts**; the GKD authors found more of that works better. **Check the teacher's licence**: some API terms forbid using outputs to train competing models. And evaluate the student on **your eval set** (Issue #26), not the teacher's benchmarks.

🔗 GO DEEPER

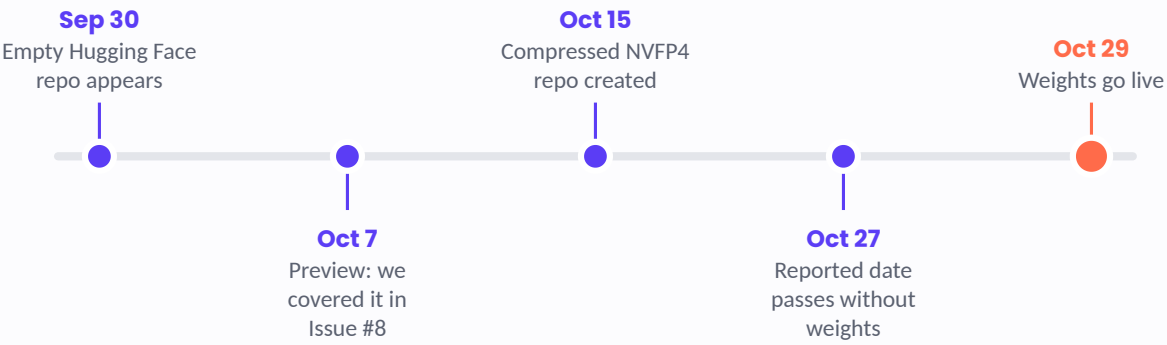
- **Distilling step-by-step** (Google Research). Why teaching the reasons, not just answers, lets tiny students win.
- **Prune and distill Llama 3.1 8B to 4B** (NVIDIA). A real recipe; NVIDIA says it needs **up to 40x fewer training tokens** than starting from scratch.
- **GKD Trainer** (Hugging Face TRL). On-policy distillation with the full runnable example.
- **Knowledge distillation** (Google MaxText). Offline and online recipes, with the settings explained.

PART 2

What moved in AI

7 min read · Covering Thursday. A trillion-parameter model you can download, browsers inside coding tools that aren't real browsers, and OpenAI's new rulebook.

Mistral Large 4: from preview to weights

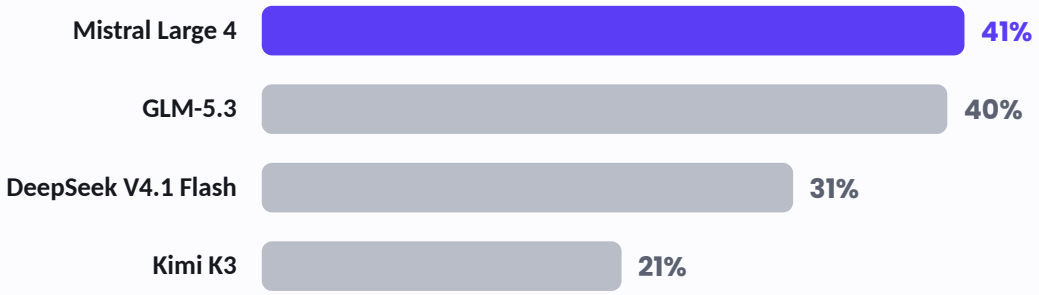


Top story 1 · Mistral opens up its trillion-parameter model

OPEN WEIGHTS CODING LICENCE

What happened. Mistral published the weights for **Mistral Large 4**, nicknamed "Le Chonk", on Hugging Face. It's a **mixture-of-experts** model of about **1.05 trillion parameters**, but only about **49B are active per token** (52B counting embeddings), which keeps it cheaper to run. It reads text and images and writes text and code. (OrcaRouter)

Terminal-Bench 4.0 (Mistral's chart)



Company-reported, via TNW. No third party has re-run the released checkpoint yet (OrcaRouter).

The details that matter:

- **Better than the preview:** Terminal-Bench went from **28% to 41%** after three extra weeks of RL training, Mistral says. It claims **47% fewer tokens per task** than GLM-5.3 on coding agents. (TNW)
- **Big download:** the main repo is **1,060.6 GB** and ungated. A compressed **NVFP4** version is **610 GB** and, per Mistral, runs on **four Nvidia B200 or B300 GPUs**.
- **The licence catch:** you can use, modify, fine-tune and sell products built on it. But if you run a **"Model as a Service" business**, renting the model out to others, you need a **separate commercial licence**. Internal use and features built into your own product are exempt.

- **API price:** a launch promo of about **$0.70 input / $2.10 output** per million tokens; OrcaRouter lists the full price at roughly double that. Context is listed at up to **1M tokens**.

Why it matters. Mistral's science VP pitched it simply: once downloaded, the weights **can't be cut off** by anyone. For most readers, the real prize is what comes next: **distilled versions small enough to run locally**.

WHO SHOULD CARE 🙋 Everyone ●●○○○ 🎨 Vibe coders ●●●○○ 🛠️ Developers ●●●●○
🔥 Hype level ●●●○○

👀 **Should you care?** You won't run this at home: it needs datacentre GPUs. But you can **try it cheaply through Mistral's API** as a coding model today. Worth benchmarking against what you pay for now.

🛠️ For devs

- Repos: `mistralai/Mistral-Large-4-1T-A52B` and `...-NVFP4`. API model: `mistral-large-4`. Day-one support in **vLLM, SGLang, llama.cpp and transformers** is reported. (ShiftDelete)
- Building an API or gateway on top? **Read the MaaS clause** before you launch; the boundary isn't fully clear yet. (itmunch)

Top story 2 · The browser inside your coding agent isn't a real browser

SECURITY FOR DEVS UNPATCHED

What happened. Gal Weizman of Forever Security tested the **built-in browsers** in Cursor, GitHub Copilot, OpenAI Codex, VS Code and Claude Desktop, the little preview panes where your agent opens web pages and local HTML files. The research, called **EMBRACE**, found every one broke **at least one rule a normal browser enforces**. (Help Net Security)

Tool	What the attack did	Status
Cursor	A page hid a command; one approval ran it, and **the agent switched off its own sandbox** to retry	Patched Oct 1
GitHub Copilot	A page could **fake the address bar**, e.g. a phishing page shown as github.com/login	Fixed in 1.1.29
Claude Desktop	A local HTML file could **read files outside the project**	Second report accepted by Anthropic
VS Code	Same file-reading trick	**Unpatched**; Microsoft declined
OpenAI Codex	Reads any file; no folder limit at all	**Unpatched**; no reply since July

The details that matter:

- **What leaked in tests: SSH keys, .env files, npm tokens** and the hosts file. (Forever Security)

- **The trigger is ordinary:** a project's README says "set up", the assistant offers a link to an HTML file, **you click once**.
- **Vendors disagree:** Microsoft and, at first, Anthropic said "Workspace Trust" (the "do you trust this folder?" prompt) is the protection. Weizman says that prompt gives **"zero protection."**

Why it matters. Vibe coders clone strangers' repos all the time, and **the agent itself draws the link you click**. Your agent's browser pane has your files, but none of the decades of protections Chrome has.

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🔥 Hype level ●○○○○

> 👀 **Should you care?** Yes, if you use any of these. Until fixes land, **don't open HTML files from projects you didn't write** inside the tool. Open them in your real browser, or not at all.

> 🛠️ **For devs: a 5-minute check**
>
> - List the folders you've marked **trusted** in VS Code or Codex and remove anything you cloned and don't need.
> - Keep secrets out of plain files where you can: a **password manager or secrets CLI** beats `.env` on disk. Rotate keys if you've opened unknown repos.

Top story 3 · OpenAI rewrites its rulebook around five values

POLICY HOW MODELS BEHAVE

What happened. OpenAI published a new version of its **Model Spec**, the public document describing how its models should behave. It's now built around **five values**: faithfulness to human intent, minimising harm, honesty and transparency, human agency, and care. **Faithfulness comes first**: grasp what people actually want, not just the literal words. (TNW)

📘 August Spec	VS	📗 October Spec
Calls the model **"the assistant"**		Calls the model **"GPT"**
Lots of **detailed style rules**		Most style rules **removed**
81 headings		70 headings, **five values** on top

Heading counts are FourWeekMBA's own comparison of the two versions, not OpenAI figures.

The details that matter:

- **Defaults, not commandments:** most rules are now **defaults that users and developers can change**, with firm limits only where needed to protect people.
- **New sections** on using **memory and past chats**, respecting people's own choices, and following social norms.

- **OpenAI's own admission:** its models **"do not yet fully reflect"** the Spec, and it is testing new training methods. It's asking the public for feedback. (FourWeekMBA)

Why it matters. It's a shift from rules to **judgment**, since rulebooks can't cover every action an agent takes. After yesterday's agent-steering study, note that **"human agency"** made the five.

WHO SHOULD CARE 🙋 Everyone ●●●○○ 🎨 Vibe coders ●●○○○ 🛠️ Developers ●●●○○

🔥 Hype level ●●○○○

⚡ Quick hits

💸 ChatGPT Pro 200 limits halve today.

The legacy grace period ended 29 Oct: Pro 200 drops from **20x to 10x** Plus usage. Existing subscribers got **62,500 credits**, valid until 31 Dec. (Startup Fortune)

🔒 Reddit goes logged-in.

Update to Issue #3: citing AI scraping and licensing, Reddit will mostly **require sign-in**, testing outside the US this quarter. Search referrals fell by **over 1M a week** in October. (TechCrunch)

🎙️ Deepgram buys Daily, maker of Pipecat.

The popular voice-agent framework **stays open source**. The two also announced an open protocol for voice agents. Price not disclosed. (TechCrunch)

🇨🇦 Cohere: up to $5B at $20.5B.

First close of its Series E; Canada's government committed up to **$1.14B**. Cohere claims ARR grew **165%**. (AI Weekly)

📊 Anthropic shows bosses the ROI.

New reports show what a team's Claude spend produced, plus **pooled token budgets**. A Jellyfish report says Anthropic tools now make **nearly a third** of proposed code changes. (TechCrunch)

🗓️ Coming up.

Reddit API stops new requests **31 Oct**. DeepSeek free window ends **1 Nov**. US midterms **3 Nov**. Kimi K3.1 Pro weights by **4 Nov**. India's draft AI rules expected **in November**. **OpenAI fine-tuning:** no new jobs after **6 Jan**.

🎯 One thing to try today: audit your agent's browser

Open your coding tool's **trusted folders** list and remove any you don't recognise. Then hunt for stray `.env` files: **what an HTML file can read, an attacker can too.**

🧠 Pop quiz: 30 seconds, no Googling

1. In distillation, the small model learns mainly from...

- A Random web pages
- B A bigger teacher model
- C User ratings

2. Mistral Large 4's licence needs a separate deal if you...

- A Use it internally
- B Fine-tune it
- C Rent it out as a model service

3. Which tools were still unpatched for EMBRACE?

- A Codex and VS Code
- B Cursor and Copilot
- C None

Answers (no peeking): 1. B: it copies the teacher's answers or scores · 2. C: the Model-as-a-Service clause · 3. A: as of 29 Oct

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